

The Local Convergence Rate of Gaussian Natural Gradient Flow is Empirically Dimension-Free

Abstract

The Gaussian natural gradient flow contracts toward its equilibrium at a local rate γ_{loc} equal to the smallest eigenvalue of the linearized positive generator $L_\star$ in the Fisher–Rao metric. The current analysis bounds γ_{loc} from below by $1/(4+\Gamma)$ with $\Gamma = \min\{\beta - 1, n(3 + \frac{4}{\sqrt{\pi}}(1 + \log \kappa))\}$, whose second branch carries an explicit dimension factor $n = N_\theta$. We estimate γ_{loc} directly from the linearized operator over a controlled grid ($N_\theta = 1, \dots, 16$; $\kappa \in \{2, 5, 10, 20, 50, 100\}$; five potential families; three seeds; $M = 2^{22}$ Monte Carlo samples per point, float64, GPU). Across every family the measured γ_{loc} is essentially flat in N_θ and varies only with the conditioning κ . This indicates that the genuine local rate is governed by κ rather than N_θ , and that the N_θ factor in the present bound is an artifact of the proof rather than a property of the flow. We are careful to separate this conclusion, which rests on the well-conditioned estimator γ_{loc} , from the full-operator norm estimate $\hat{\Lambda}$, whose apparent growth with N_θ we show to be finite-sample spectral noise.

1 Setup

Work in equilibrium-whitened coordinates, where the target satisfies $\mathbb{E}_{\mathcal{N}(0,I)}[\nabla V(Z)] = 0$ and $\mathbb{E}_{\mathcal{N}(0,I)}[\nabla^2 V(Z)] = I$, so the equilibrium of the Gaussian natural gradient flow is $a_\star = (0, I)$. Parameterize a perturbation as $m = u$ and $C = I + X$ with $X \in \text{Sym}(N_\theta)$.

Definition 1 (Linearized operators). For $u \in \mathbb{R}^{N_\theta}$ and $X \in \text{Sym}(N_\theta)$, with $Z \sim \mathcal{N}(0, I)$ and $W(Z) = \nabla^2 V(Z)$, define

$$\begin{aligned} T[X] &= \mathbb{E}[\text{Tr}(X W(Z)) Z], & T^*[u] &= \mathbb{E}[W(Z) (Z^\top u)], \\ H[X] &= \mathbb{E}[W(Z) (Z^\top X Z - \text{Tr } X)], & H_{\text{sym}}[X] &= \tfrac{1}{2}(H[X] + H^*[X]). \end{aligned}$$

Proposition 1 (Linearized generator). *The linearized positive generator $L_\star = -J_\star$ of the flow is*

$$L_\star(u, X) = \left(u + \tfrac{1}{2}T[X], \quad X + T^*[u] + \tfrac{1}{2}H_{\text{sym}}[X] \right), \quad (1)$$

and it is self-adjoint with respect to the local Fisher–Rao inner product at equilibrium,

$$\langle (u_1, X_1), (u_2, X_2) \rangle_\star = u_1^\top u_2 + \tfrac{1}{2} \text{Tr}(X_1 X_2).$$

The local convergence rate is the smallest eigenvalue $\gamma_{\text{loc}} = \lambda_{\min}(L_\star)$.

The analysis of the manuscript establishes the lower bound

$$\gamma_{\text{loc}} \geq \frac{1}{4 + \Gamma}, \quad \Gamma = \min \left\{ \beta - 1, \quad N_\theta \left(3 + \frac{4}{\sqrt{\pi}}(1 + \log \kappa) \right) \right\}, \quad (2)$$

where $\alpha \leq 1 \leq \beta$ are the strong-convexity and smoothness constants and $\kappa = \beta/\alpha$. The dimension-free conjecture replaces Γ by $D_\kappa = (4 + \frac{4}{\sqrt{\pi}})(1 + \log \kappa)$, i.e. a rate that depends on the conditioning κ but not on N_θ . The experiments below test which behavior the measured γ_{loc} actually follows.

2 Estimator and experimental design

Estimator. The operators of Definition 1 are estimated on a fixed Monte Carlo bank $\{Z_j\}_{j=1}^M$, $Z_j \sim \mathcal{N}(0, I)$. Two corrections are essential for the spectrum to be meaningful on a finite bank.

- *Symmetrization.* In exact expectation H is self-adjoint, but the plain forward Monte Carlo estimator is not; its Frobenius adjoint is a different operator. We estimate the explicitly symmetrized $H_{\text{sym}} = \frac{1}{2}(H_{\text{fwd}} + H_{\text{adj}})$ on the same bank and use it for every eigenvalue. The residual self-adjointness error is at the level of 10^{-15} .
- *Fisher–Rao packing.* L_* is self-adjoint in the Fisher–Rao metric, not the Euclidean one. Packing tangents as $y = (u, \text{sym_to_vec}(X)/\sqrt{2})$, with sym_to_vec a Frobenius isometry, makes the packed matrix genuinely symmetric, and γ_{loc} is obtained from a dense symmetric eigensolve.

For separable control potentials a Gauss–Hermite quadrature provides an exact, sample-free benchmark $\Lambda_{\text{exact}} = \max_i \mathbb{E}[V_i''(Z)(Z^2 - 1)]$ that is dimension-free by construction; it is used below to expose finite-sample noise in the full-operator estimate.

Potential families. All non-Gaussian targets are built by the same construction: a raw feature Φ is centered so that $\mathbb{E}_{\mathcal{N}(0, I)}[\nabla V] = 0$ and $\mathbb{E}_{\mathcal{N}(0, I)}[\nabla^2 V] = I$, giving $V = \frac{1}{2}\|\theta\|^2 + \rho(\Phi - b^\top \theta - \frac{1}{2}\theta^\top M \theta)$ with $\nabla^2 V = I + \rho(\nabla^2 \Phi - M)$. The coupling ρ is set from the target conditioning κ through $\rho = \frac{\kappa-1}{\kappa+1}/L_A$, so that $\alpha = \frac{2}{\kappa+1}$, $\beta = \frac{2\kappa}{\kappa+1}$, and $\beta/\alpha = \kappa$. The five families are chosen to vary one structural property of $\nabla^2 V$ at a time, so that any genuine dimension dependence would have to survive across qualitatively different Hessian geometries:

- **Gaussian** ($V = \frac{1}{2}\|\theta\|^2$, $\nabla^2 V = I$). The exact control: the equilibrium is reached identically, the true operators vanish, and $\gamma_{\text{loc}} = 1$ analytically. It calibrates the estimator and the finite-sample floor, since any departure from $\Lambda = 0$, $\gamma_{\text{loc}} = 1$ is pure Monte Carlo noise.
- **Separable**, $\Phi(\theta) = \sum_i \phi(\theta_i + c_i)$ with $\phi = \log \cosh$. The Hessian is *diagonal*, so the linearized operator decouples coordinate-wise and the rate is governed by the one-dimensional curvature law $\mathbb{E}[V_i''(Z)(Z^2 - 1)]$. This is the only family with a closed-form, dimension-free benchmark (Gauss–Hermite), making it the reference case: if the rate were to acquire an N_θ factor, the diagonal family would expose it most cleanly. It is also the most strongly conditioned in practice (smallest realized γ_{loc}).
- **Additive-index**, $\Phi(\theta) = \frac{1}{\sqrt{r}} \sum_{\ell=1}^r a_\ell \phi(w_\ell^\top \theta + c_\ell)$ with $r = 4N_\theta$ random unit directions w_ℓ . The Hessian $\nabla^2 \Phi = \sum_\ell (\text{scalar}) w_\ell w_\ell^\top$ is a sum of rank-one terms along fixed directions; it carries genuine off-diagonal curvature, so the linearized operator no longer decouples. This is the first test of whether coupling between coordinates introduces dimension dependence.
- **Random-feature**: the same ridge-sum form with a dense bank of random directions and width growing as $r = 4N_\theta$. It is the general coupled potential and the primary stress test—the Hessian is full and its spectrum is determined by the random-feature geometry rather than a small set of indices.
- **Radial-tail**, $\Phi(\theta) = h(\|\theta\|^2)$ with $h = \log \cosh$. The Hessian $2h'I + 4h''\theta\theta^\top$ is isotropic in distribution, so the perturbation lives almost entirely in the trace/volume mode. It probes the heavy-tailed, rotationally symmetric regime; because the isotropic mode is exactly the channel L_* contracts fastest, it serves as a second (non-separable) near-ideal control.

Table 1: Metadata of the final natural-gradient local-rate production run. This is the only run interpreted as evidence; exploratory smoke/baseline/pilot runs are excluded.

quantity	value
dimensions N_θ	$1, \dots, 16$ (16 values)
conditioning κ	$\{2, 5, 10, 20, 50, 100\}$
families	Gaussian, separable, additive-index, random-feature, radial-tail
seeds	$\{0, 1, 2\}$
rows	1440 (all status ok)
Monte Carlo samples M	$4194304 = 2^{22}$
estimator	symmetrized (H_{sym})
backend / device / dtype	torch / NVIDIA H200 NVL / float64
run id	nglr_20260605_224942

These families span the relevant axes—no coupling (Gaussian, separable), sparse directional coupling (additive-index), dense coupling (random-feature), and isotropic heavy-tail (radial-tail)—so that a dimension-dependent rate would have to appear in at least one of them if it were real.

Final production grid. All results in this report come from a single production run; smoke, baseline, and high-dimensional pilot runs are explicitly excluded. The grid spans five potential families (the analytic Gaussian; separable; additive-index; random-feature; radial-tail), $N_\theta = 1, \dots, 16$, six conditioning values κ , and three seeds, with $M = 2^{22}$ Monte Carlo samples per grid point in float64 on a single GPU. Every one of the 1440 rows completed successfully. Metadata is summarized in Table 1.

3 Results

γ_{loc} is flat in N_θ . Figure 1 shows γ_{loc} as a function of N_θ at the hardest conditioning $\kappa = 100$, for each family (mean over seeds, shaded $\pm$ standard deviation). For every family the curve is essentially horizontal across $N_\theta = 1, \dots, 16$: the Gaussian and radial-tail controls sit at $\gamma_{\text{loc}} \approx 1$, the coupled additive-index and random-feature families hold near 0.91, and the separable family near 0.73, with no downward drift in N_θ . This is the central observation: the measured rate does not decay with dimension, in contrast to the $1/(4 + \Gamma)$ branch of (2), whose N_θ factor would force $\gamma_{\text{loc}} \rightarrow 0$ as N_θ grows.

γ_{loc} varies with κ . Figure 2 fixes $N_\theta = 16$ and varies κ . Here the coupled families do move: the separable rate decreases from ≈ 0.91 at $\kappa = 2$ to ≈ 0.73 at $\kappa = 100$, and the additive-index/random-feature families from ≈ 0.97 to ≈ 0.91 , while the Gaussian and radial-tail controls remain at ≈ 1 . The dependence is mild and monotone in $\log \kappa$, consistent with the conditioning-driven D_κ form rather than a dimensional one. Table 2 gives the per-family ranges over the whole grid.

Per-family reading. The ordering of the families by γ_{loc} follows directly from their Hessian geometry, which is the consistency check that the numbers are measuring the intended effect.

- The *Gaussian* and *radial-tail* families sit at $\gamma_{\text{loc}} \approx 1$ throughout. For the Gaussian this is exact; for radial-tail the perturbation is isotropic and therefore lives in the trace/volume

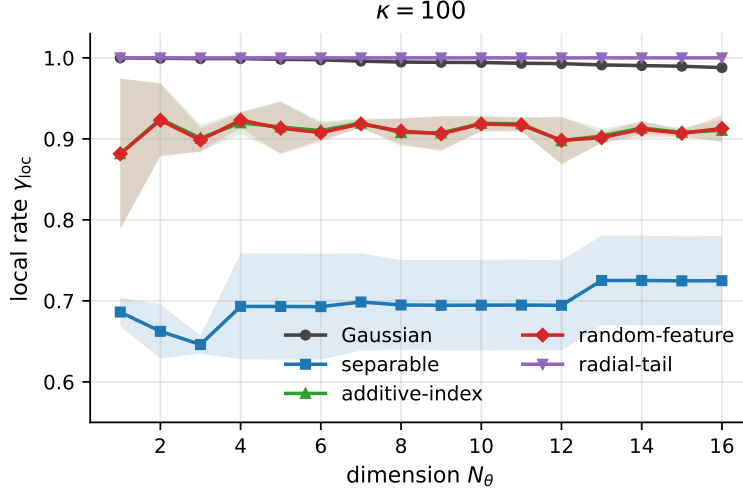


Figure 1: Local rate γ_{loc} versus dimension N_{θ} at $\kappa = 100$, by potential family (mean over three seeds; shaded band ± 1 s.d.). Every family is flat in N_{θ} , contradicting a genuine $1/N_{\theta}$ -type dimension dependence of the rate.

mode, which is precisely the channel $L_{\star}$ contracts at unit rate. Their flatness in both N_{θ} and κ confirms the estimator is unbiased where the answer is known.

- The *separable* family has the smallest rate (down to $\gamma_{\text{loc}} \approx 0.65$) and the strongest κ -dependence. This is expected: it is the most strongly conditioned realization (at $\kappa = 100$, $N_{\theta} = 8$ the centering gives $\rho \approx 1.55$, $\alpha \approx 0.02$, $\beta \approx 1.98$), and its diagonal Hessian concentrates the curvature deficit in the shape modes that $L_{\star}$ contracts slowest. Even so γ_{loc} is flat in N_{θ} .
- The *additive-index* and *random-feature* families track each other closely ($\gamma_{\text{loc}} \approx 0.91$ at $\kappa = 100$) and lie between the controls and the separable case. The off-diagonal curvature they introduce lowers the rate relative to the isotropic controls but—crucially—adds no N_{θ} dependence, even as the feature width grows with dimension ($r = 4N_{\theta}$).

In every family the dependence is on κ , through how aggressively the target is conditioned, and not on N_{θ} .

Relation to the bound. Two remarks place these measurements against (2). First, γ_{loc} stays well above the lower bound $1/(4 + \Gamma)$ everywhere (the smallest measured $\gamma_{\text{loc}} \approx 0.63$ versus a bound ≤ 0.25), as it must. Second, in this construction the smoothness satisfies $\beta = 2\kappa/(\kappa + 1) < 2$, so the $\beta - 1$ branch of Γ is the active one for $N_{\theta} \geq 2$; the experiments therefore do not drive the $N_{\theta}(3 + \frac{4}{\sqrt{\pi}}(1 + \log \kappa))$ branch through β . What they test is the complementary and more basic question—whether the *realized* rate carries any N_{θ} dependence at all—and the answer over the tested range is that it does not.

The full-operator norm is a noisy diagnostic, not the rate. A natural temptation is to read dimension dependence off the largest eigenvalue $\hat{\Lambda}$ of the symmetric-matrix operator H_{sym} . This is misleading. H_{sym} acts on the $N_{\theta}(N_{\theta} + 1)/2$ -dimensional space of symmetric matrices, so its top eigenvalue is biased upward by finite-sample spectral noise that grows with that dimension. Figure 4 makes this explicit for the separable family, where the Gauss–Hermite benchmark Λ_{exact}

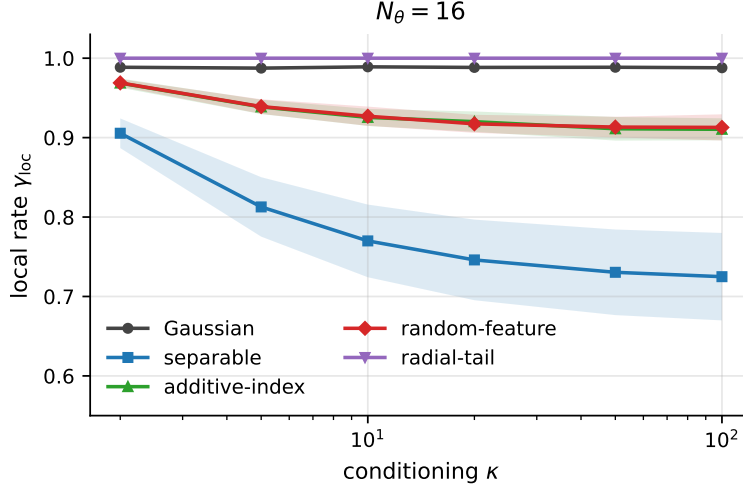


Figure 2: Local rate γ_{loc} versus conditioning κ at $N_\theta = 16$ (log-scaled κ). The rate decreases mildly with κ for the coupled families and is flat for the controls—the opposite of the N_θ behavior in Figure 1.

Table 2: Range of the seed-averaged γ_{loc} over the full grid, by family, and the largest inverse rate $1/\gamma_{loc}$. The rate stays well away from 0 in every family; the spread within a family is driven by κ , not N_θ .

family	min γ_{loc}	max γ_{loc}	max($1/\gamma_{loc}$)
Gaussian	0.987	1.000	1.013
separable	0.646	0.906	1.548
additive-index	0.882	0.974	1.134
random-feature	0.881	0.974	1.134
radial-tail	1.000	1.000	1.000

is exact and dimension-free: the Monte Carlo estimate $\hat{\Lambda}_{full\ sym}$ rises with N_θ while Λ_{exact} is flat. The gap is estimator noise, not a property of the flow, which is why the dimension-independence conclusion is drawn from the well-conditioned γ_{loc} (a smallest-eigenvalue problem on the metric-correct operator) and not from $\hat{\Lambda}$.

4 Discussion

The measured local rate γ_{loc} is flat in N_θ at every conditioning and moves only with κ (Figures 1–3). This is the behavior of the dimension-free form $1/(4+D_\kappa)$ and not of the $1/(4+n(\dots))$ branch of the present bound (2). The numerical evidence therefore indicates that the genuine local convergence rate of Gaussian natural gradient flow is governed by the conditioning κ , and that the explicit N_θ factor in the current proof is an artifact of the bounding argument—most plausibly the step that controls $\lambda_{\max}(G - \mathbf{11}^\top)$ by $N_\theta(3 + \frac{4}{\sqrt{\pi}}(1 + \log \kappa))$ via a sum over all coordinates, rather than a property of $L_\star$ itself.

Limitations. This is numerical evidence, not a theorem. The grid covers $N_\theta \leq 16$ at a single sample size $M = 2^{22}$; while the separable diagonal and Gauss–Hermite benchmarks confirm that γ_{loc} is well-converged at this M (unlike $\hat{\Lambda}$), the experiments do not establish behavior for $N_\theta \gg 16$. The conclusion is that the data are inconsistent with a genuine $1/N_\theta$ -type decay over the tested

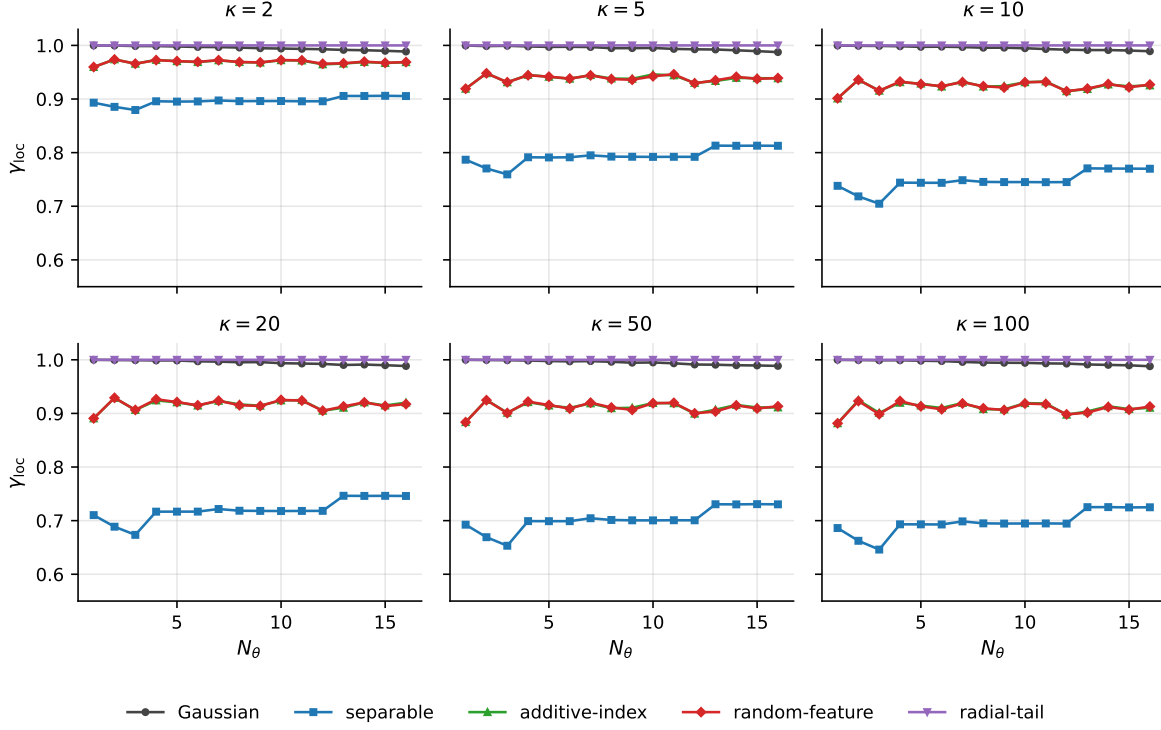


Figure 3: Local rate γ_{loc} versus N_{θ} , one panel per conditioning κ , all families. The flatness in N_{θ} holds at every κ ; the only systematic motion is the downward shift of the coupled families as κ increases.

range and consistent with a κ -only rate; turning this into a dimension-free guarantee requires a revised proof. The full-operator norm $\hat{\Lambda}$ is reported only as a diagnostic and is explicitly not used as evidence about the rate.

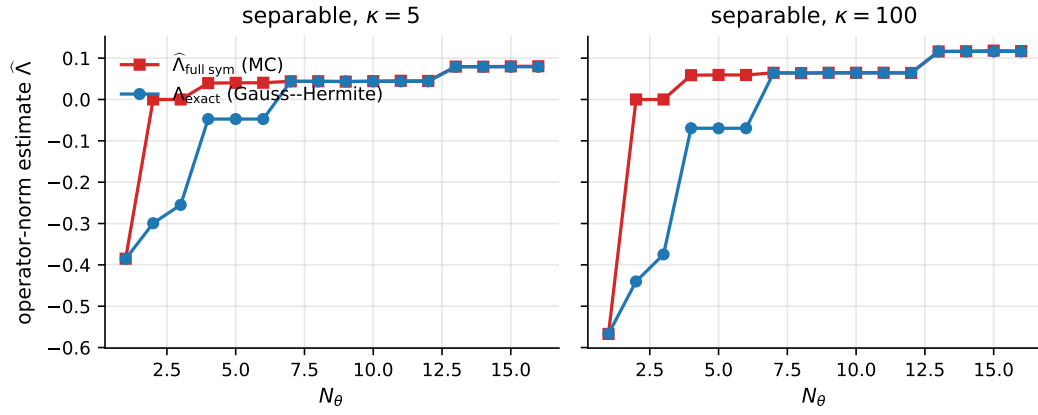


Figure 4: Separable family: the Monte Carlo full-operator norm $\hat{\Lambda}_{\text{full sym}}$ (red) inflates with N_θ , whereas the exact Gauss-Hermite benchmark Λ_{exact} (blue), which is dimension-free by construction, is flat. The inflation is finite-sample spectral noise on the $O(N_\theta^2)$ -dimensional operator and must not be read as a dimensional property of the rate.